Global Arms Transfers and Defense Trade Architecture (2000–2024)

A Quantitative Analysis of Supplier Concentration, Decadal Geopolitical Shifts, and Regional Defense Balances

Project Report

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Abstract

This project examines the structural dynamics of conventional international defense trade between 2000 and 2024, focusing on supplier concentration, the presence of structural breaks in the wake of the geopolitical shift around 2014, the evolution of trade volumes over time, and trade imbalances across regions. The data sources are the Stockholm International Peace Research Institute (SIPRI) Trend Indicator Values (TIV) dataset, with spreadsheet data cleaning (Microsoft Excel), relational data modeling (MySQL 8.0), statistical validation and exploratory charting (Python), and dashboard visualization (Microsoft Power BI).

The original data consists of 3,985 total observations, with 2,973 observations from sovereign states in 181 different countries (excluding the regional macro-aggregates), from 1985-2010. Over the total period, 646.33 billion TIV were exported globally and 636.33 billion TIV were imported. The results reveal high supplier concentration, the top two suppliers (34.4% and 20.8% respectively, the United States and the Russian Federation) share 55.2% of all cumulative global exports and the top 10 exporters 88.1% of all cumulative global exports. In line with the international strategic changes that took place after the 2014 annexation of Crimea, Russian exports decreased by 42.65%, whereas US exports increased by 19.83%, French exports by 26.28% and exports from the emerging middle power countries such as South Korea (+175.90%) and Türkiye (+300.92%) grew rapidly. Structurally, North America and Europe are net exporters, whereas the Middle East & North Africa and South Asia are net importers.

The project demonstrates quantitative data analysis methods for empirical questions in three areas: political science, defense trade policy and international political economy. The results are descriptive and do not show causal determinism.

1. Introduction

The sale of the major conventional armaments is an integral part of the global distribution of power, power treaties and foreign policy alignment. In international relations, arms transfer studies can be used to explain the military-industrial capability of the supplier and recipient states, as well as the relationship between them and changes in regional security complexes. Arms transfers were not just in monetary terms, but in volume indices, which reflect the real number of deliveries of military equipment, such as aircraft, armored vehicles, missile defense systems, radar installations, naval vessels, etc.

The traditional arms industry is more than an arms deal. It includes long-term maintenance, ammunition supply and technology licensing, and influences trade balances of recipient countries.

These transfers can be quantified over time to gain empirical knowledge of trade polarity, the reliability of trade partners, and the effect of geopolitical events on trade volumes.

This project seeks to look at indicators of conventional arms trading between sovereign states from 2000 to 2024 based on data gathered by SIPRI and the World Bank. The project consists of data ingestion, restructuring of the tabular data, SQL database design, Python statistical scripting, and reporting through an interactive Power BI dashboard to analyze market concentration, strategic timeline changes, and regional defense balance differences.

2. Research Objectives

The project has five main objectives:

- Measure cumulative global export volume and calculate the market share of the top global arms suppliers between 2000 and 2024.
- Evaluate supplier concentration using cumulative export shares and identify the primary arms-producing nations.
- Examine structural shifts in exports and imports before and after the 2014 geopolitical inflection point (comparing 2000–2013 against 2014–2024).
- Analyze the longitudinal year-over-year trajectory of global defense trade volumes across the 25-year timeframe.
- Compare regional trade balances, net procurement deficits, and export-to-import ratios across World Bank regional classifications.

3. Data Source and Scope

The project uses the World Development Indicators (WDI) defense trade series published by the World Bank, utilizing raw indicator values established by the Stockholm International Peace Research Institute (SIPRI).

- **Indicator Codes:** MS.MIL.XPRT.KD (Arms exports in SIPRI TIV) and MS.MIL.MPRT.KD (Arms imports in SIPRI TIV).
- **Measurement Unit:** SIPRI Trend Indicator Value (TIV), expressed in constant 1990 US dollars (in millions/billions), representing the volume of military resources transferred rather than raw financial transaction values. Deliveries below 0.5 million TIV are officially recorded as 0.

The dataset scope comprises:

- **Time Span:** 2000 to 2024 (25 annual intervals).
- **Total Records:** 3,985 rows across sovereign states and regional groupings.
- **Analytical Scope:** 2,973 valid records covering 181 sovereign states. Macro-regional aggregates (e.g., 'World', 'Sub-Saharan Africa (excluding high income)', 'Arab World') were excluded from national-level rankings to prevent double-counting.

4. Data Preparation and Methodology

4.1 Data Cleaning

The raw data originally came in wide-format longitudinal tables with individual years spanning columns 1960 through 2024. Data cleaning was performed as follows:

- Unpivoting yearly columns into tabular rows with schema (Country_Name, Country_Code, Year, Imports_TIV, Exports_TIV).
- Standardizing country codes to three-letter ISO-3 formats.
- Converting formatted string figures containing thousand-separators into integer data types and treating empty records or values below reporting thresholds as zero (0).
- Calculating row-level derived attributes: Net Trade TIV = Exports TIV - Imports TIV, and Total Volume TIV = Exports TIV + Imports TIV.
- Merging World Bank country metadata tables to incorporate regional classifications (Region) and income groups (Income_Group).

4.2 Analytical Approach

The analytical workflow combined four integrated tools:

- **Microsoft Excel:** Initial data structuring, wide-to-long transformation, and workbook navigation documentation.
- **MySQL 8.0:** Creating a relational fact table (fact_arms_transfers), index creation on (Country_Code, Year) and (Region), and querying analytical metrics using Common Table Expressions (CTEs), DENSE_RANK(), LAG() window functions, and epoch-based conditional sums.
- **Python (Pandas & NumPy):** Data integrity verification, descriptive statistics, group aggregations, and validation of cumulative percentage shares.
- **Matplotlib & Seaborn:** Generating publication-style bar charts and longitudinal trend plots.
- **Microsoft Power BI:** Data modeling, authoring DAX measures, configuring dynamic temporal/regional slicers, and building a multi-visual executive overview dashboard.

The findings presented below represent descriptive empirical calculations from the dataset; they do not establish causal relationships or statistical significance tests.

5. Key Findings

5.1 Top Exporters and Market Concentration (2000–2024)

Cumulative arms exports from 2000 to 2024 totaled 646,334,000,000 TIV globally across all sovereign states. Export capacity is heavily concentrated among a small number of industrial suppliers.

Rank	Country	Cumulative Exports (TIV)	Market Share (%)	Cumulative Share (%)
1	United States	222,364,000,000	34.4%	34.4%
2	Russian Federation	134,441,000,000	20.8%	55.2%
3	France	50,421,000,000	7.8%	63.0%
4	Germany	41,813,000,000	6.5%	69.5%
5	China	30,914,000,000	4.8%	74.3%
6	United Kingdom	27,483,000,000	4.3%	78.6%
7	Italy	18,116,000,000	2.8%	81.4%
8	Israel	16,890,000,000	2.6%	84.0%
9	Spain	13,942,000,000	2.2%	86.2%
10	Netherlands	12,579,000,000	1.9%	88.1%

The top two exporters accounted for more than half (55.2%) of all major conventional weapons delivered over the past 25 years. The top 5 suppliers accounted for 74.3%, and the top 10 accounted for 88.1%, indicating an oligopolistic market structure in global defense manufacturing.

5.2 Strategic Decadal Shift: Pre-2014 vs. Post-2014

To examine shifts across major geopolitical periods, the dataset was divided into two eras: Era 1 (2000–2013) and Era 2 (2014–2024). The year 2014 marks an important inflection point in international defense procurement, coinciding with the annexation of Crimea and subsequent defense spending reassessments across NATO and non-NATO states.

Country	Pre-2014 Exports	Post-2014 Exports	Post-2014 Trend

	(TIV)	(TIV)	Growth (%)	Direction
United States	101,152,000,000	121,212,000,000	+19.83%	Increase (↑)
Russian Federation	84,402,000,000	50,039,000,000	-42.65%	Decrease (↓)
France	22,282,000,000	28,139,000,000	+26.28%	Increase (↑)
Germany	23,678,000,000	18,135,000,000	-23.41%	Decrease (↓)
China	12,187,000,000	18,727,000,000	+53.66%	Increase (↑)
United Kingdom	15,317,000,000	12,166,000,000	-20.57%	Decrease (↓)
Korea, Rep.	2,231,000,000	6,689,000,000	+199.82%	Increase (↑)
Türkiye	1,226,000,000	3,688,000,000	+200.82%	Increase (↑)

Russian exports experienced a substantial drop (-42.65%) in the post-2014 period, reflecting the impact of international trade sanctions, internal procurement needs, and supply chain adjustments. In contrast, the United States (+19.83%) and France (+26.28%) expanded their export volumes. Mid-tier suppliers such as South Korea and Türkiye recorded significant percentage increases, pointing to diversification in global supply networks.

5.3 Longitudinal Trade Trajectory

Examining the annual volume trajectory between 2000 and 2024 shows cyclical patterns in global defense deliveries:

- **2000–2004:** A relatively steady baseline with annual global export volumes averaging between 17 and 21 billion TIV.
- **2005–2012:** A steady upward climb, peaking above 29 billion TIV around 2011–2012.
- **2013–2018:** A sustained plateau characterized by elevated delivery volumes, reaching an overall annual peak of 33.3 billion TIV in 2021.
- **2022–2024:** A period of divergence where deliveries settled around 29.4 to 30.7 billion TIV annually, accompanied by changing supplier-recipient arrangements.

5.4 Regional Trade Balances and Dependency

Aggregating defense transfers across the seven World Bank geographic regions reveals pronounced differences between net suppliers and net procurement zones:

Geographic Region	Total Exports	Total Imports	Net Balance	Export/Import

	(TIV)	(TIV)	(TIV)	Ratio
Europe & Central Asia	338,829,000,000	129,208,000,000	+209,621,000,000	2.62x
North America	227,398,000,000	23,059,000,000	+204,339,000,000	9.86x
East Asia & Pacific	41,563,000,000	127,110,000,000	-85,547,000,000	0.33x
Middle East & North Africa	22,600,000,000	196,819,000,000	-174,219,000,000	0.11x
South Asia	2,752,000,000	108,011,000,000	-105,259,000,000	0.03x
Sub-Saharan Africa	7,651,000,000	29,380,000,000	-21,729,000,000	0.26x
Latin America & Caribbean	5,541,000,000	22,544,000,000	-17,003,000,000	0.25x

North America and Europe & Central Asia are the only net-exporting regions, holding export-to-import ratios of 9.86 and 2.62 respectively. The Middle East & North Africa and South Asia exhibit large net deficits, indicating ongoing structural reliance on external defense equipment suppliers.

6. Discussion

The results indicate significant trends in global arms trade. Traditional manufacturers (mainly in North America and Western/Central Europe) continue to dominate global defense transfers. This is due to high entry barriers such as long and expensive R&D, scale and established alliance structures.

Simultaneously, it is important to observe that supplier rankings are not fixed as illustrated by the comparison between the ranking from before and after 2014. The decline in Russian conventional arms sales and the expansion of secondary arms suppliers such as South Korea and Türkiye imply a certain degree of market diversification. These countries seem to be expanding their procurement choices in order to improve their military capabilities or to reduce their reliance on a single partner.

Examples of the global variability in defense dependency are exposed in regional trade balances. Regional trade balances illustrate defense dependency in different ways around the world. Advanced economies in Europe and North America have surpluses of equipment while areas with continuing security tensions, in particular, the Middle East and South Asian nations, are taking in significant import levels without domestic export production. These imbalances are related to the different military-industrial developments and defense alignments.

7. Limitations

Several limitations should be considered when interpreting these findings:

- **Descriptive Analysis:** The study presents descriptive figures and longitudinal comparisons; it does not test causal hypotheses or assess the statistical significance of geopolitical events.
- **SIPRI TIV Unit of Measurement:** The SIPRI Trend Indicator Value measures the volume of military resource transfers rather than contract financial values. It should not be interpreted as financial balance-of-payments data.
- **Suppression of Minor Transfers:** Transfers below 0.5 million TIV are rounded to zero in reporting, which may undercount small arms transfers and low-value component shipments.
- **Exclusion of Non-State Actors:** The analysis considers sovereign states and does not include transfers to non-state groups or covert weapons shipments.
- **Macro-Aggregates:** Regional aggregations pool diverse domestic defense policies together, which may obscure distinct national choices within the same continent.

8. Conclusion

This project examined SIPRI/World Bank conventional arms transfer data for the period 2000-2024 to assess the concentration of suppliers and major geopolitical changes, decadal trends, annual trends and regional trade balances.

The findings indicate that the global arms market is highly concentrated, consisting of a few primary arms suppliers, namely more than half of deliveries between the United States and Russia during the 25 years. Since 2014, substantial changes appeared, such as a drop in the volume of the Russian exports and an increase of the exports realized by mid-tier exporters. Regional statistics reinforce the dependence of the regions, particularly South Asia and the Middle East as key net importers.

The study illustrates the potential for the integration of data organization, SQL querying, Python analysis, and Power BI visualization to explore empirical questions in political science, security studies, and international economics. Future research may combine defence expenditure, geopolitical alliance information and bilateral trade relations to further explore the causes of arms procurement.

9. Tools and Technologies

- **Microsoft Excel:** Data ingestion, column unpivoting (1NF restructuring), variable documentation, and cross-tabulations.

- **SQL (MySQL 8.0):** Relational schema design, fact table ingestion (LOAD DATA LOCAL INFILE), window functions (DENSE_RANK, LAG), CTEs, and segmented period summaries.
- **Python (Pandas & NumPy):** Data validation, aggregate verification, and metric modeling.
- **Matplotlib & Seaborn:** Generation of annotated horizontal bar charts and longitudinal trend plots.
- **Microsoft Power BI:** Data modeling, writing 7 custom DAX measures, dynamic slicer filtering, and interactive dashboard layout design.

10. Data Reference

Stockholm International Peace Research Institute (SIPRI). Arms Transfers Database: Trend Indicator Values (TIV). Accessed via the World Bank World Development Indicators (WDI) data portal:

- **Exports Indicator:** MS.MIL.XPRT.KD (Arms exports in SIPRI trend indicator values).
- **Imports Indicator:** MS.MIL.MPRT.KD (Arms imports in SIPRI trend indicator values).
- **Data Sources:** SIPRI Arms Transfers Programme (https://www.sipri.org/databases/armstransfers) and World Bank Open Data (https://data.worldbank.org/).